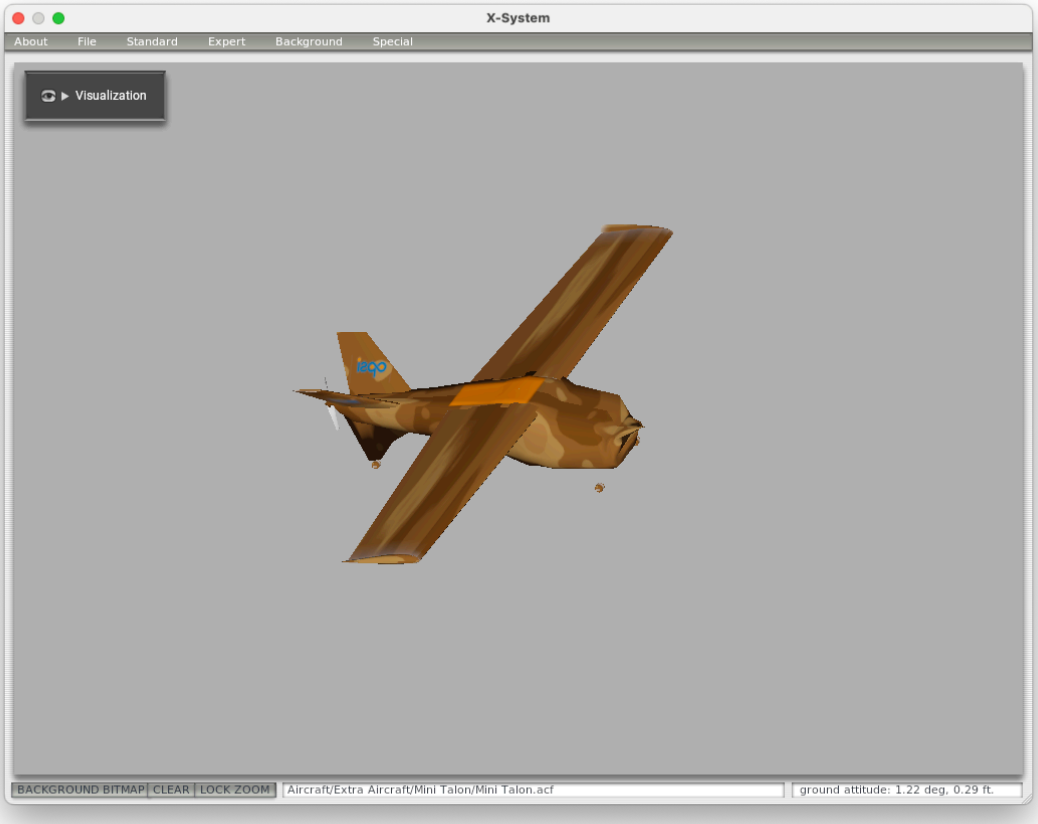
UAV Satria Talon — Spesifikasi & Evaluasi Performa

> ***Airfoil:*** *NACA 2412* · ***Semi-length:*** *2.13 ft (confirmed Plane Maker)*
> ***Empty weight:*** *3.00 lb (1.361 kg)* · ***MTOW:*** *5.25 lb (2.381 kg)*
> ***Wingspan:*** *~129.8 cm* · ***Wing area:*** *0.2731 m²* · ***AR:*** *6.17*
> ***CG updated:*** *fwd 0.91 ft · nominal 0.92 ft · aft 0.96 ft*

> ***Catatan metodologi:*** *Seluruh spesifikasi geometri, berat, dan keseimbangan disusun berdasarkan kajian desain yang dilakukan di* ***X-Plane Plane Maker*** *— simulator aerodinamis yang digunakan untuk memodelkan sayap, airfoil, CG envelope, dan parameter performa UAV Satria Talon secara iteratif sebelum divalidasi melalui uji terbang.*



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1. Ringkasan Perubahan

Before/After — Semi 1.64→2.13 ft + NACA 2412

Sebelum Generic airfoil, semi 1.64 ft	Konfigurasi	Sesudah (NACA 2412 + 2.13 ft) ★ NACA 2412, semi 2.13 ft
100 cm	Wingspan	**129.8 cm (~130 cm)**
1.640 ft	Semi-length	**2.130 ft**
0.210 m²	Wing area	**0.273 m²**
4.8	Aspect ratio	**6.17**
~1.30 (generic)	CLmax	**1.50 (NACA 2412)**
0.028	CD0 total	**0.026 (–7%)**
10.2	L/D max	**12.37 (+21%)**
11.0 m/s (39.6 km/h)	Stall speed Vs	**9.65 m/s (34.7 km/h) –12%**
17.9 m/s (64.4 km/h)	Cruise speed	**14.74 m/s (53.1 km/h)**
64 W	Cruise power	**46 W (–28%)**
95 mnt	Endurance (4S 8Ah)	**131 mnt (+38%)**
~80 km	Range one-way (4S 8Ah)	**~115 km (+44%)**
~34–36 km	RTH radius (4S 8Ah)	**~49–52 km**

2. Weight & Balance (Updated)

Weight & Balance — Satria Talon (Updated)

Empty weight	3.00 lb	1.361 kg
Maximum weight (MTOW)	5.25 lb	2.381 kg
Payload total (MTOW–empty)	2.25 lb	1.021 kg
Fuel load	0.00 lb	0 — electric platform
Long CG — forward limit	0.91 ft	26.4% MAC · 5.56 cm dari LE MAC
Long CG — default/nominal	0.92 ft	27.9% MAC · 5.87 cm dari LE MAC ✓
Long CG — aft limit	0.96 ft	33.7% MAC · 7.09 cm dari LE MAC
Vert CG	0.00 ft	—
Static margin @ nominal CG	–	~12.1% MAC (NP est. ~40% MAC) — AMAN
Static margin @ aft limit	–	~6.3% MAC — MARGINAL (jangan dilampaui)

Payload Breakdown @ MTOW 2.381 kg

Komponen	Berat	Sisa kargo
Empty weight	1.361 kg	1.021 kg

+ Baterai 4S 6000 mAh	+350 g	**671 g**
+ Baterai 4S 8000 mAh	**+450 g**	**571 g**

Analisis Perubahan CG Envelope

Titik CG	Lama	Baru	Perubahan
Forward limit	0.90 ft (25.0% MAC)	**0.91 ft (26.4% MAC)**	+0.01 ft, maju +1.4%
Default/nominal	0.95 ft (32.2% MAC)	**0.92 ft (27.9% MAC)**	−0.03 ft, mundur −4.3%
Aft limit	1.00 ft (39.5% MAC)	**0.96 ft (33.7% MAC)**	**−0.04 ft, mundur −5.8%**

> ***Mengapa aft limit diperketat ke 0.96 ft?***
> *NACA 2412 memiliki Cm_ac = −0.05 (pitching moment nose-down). Semakin CG ke belakang, momen ini semakin sulit dikompensasi — terutama di kecepatan rendah mendekati stall. Batas 0.96 ft (33.7% MAC) memberikan static margin minimum ~6.3% yang masih aman. Melampaui ini berisiko pitch-up instability.*

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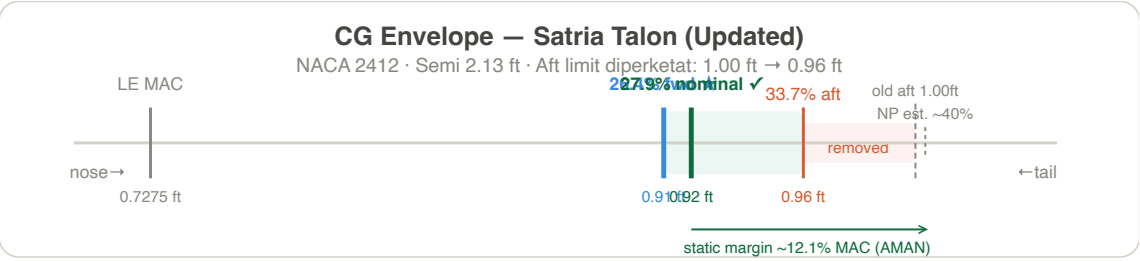
3. CG Envelope (Updated)



Titik CG	Nilai	% MAC	Dari LE MAC	Static Margin	Status
LE MAC (referensi)	0.7275 ft	0%	0 cm	—	—
Forward limit	**0.91 ft**	**26.4%**	**5.56 cm**	13.6%	AMAN
Nominal terbang	**0.92 ft**	**27.9%**	**5.87 cm**	12.1%	**AMAN ✓**
Aft limit (JANGAN dilampaui)	**0.96 ft**	**33.7%**	**7.09 cm**	6.3%	MARGINAL
Neutral Point (estimasi)	~1.001 ft	~40%	~8.4 cm	—	referensi

Cara mengukur CG fisik (updated):

1. Dari root LE, ukur ke tip sejauh **~29% semi-span** (~18.8 cm) → titik LE MAC
2. Tandai **5.56–5.87 cm ke belakang** dari LE MAC → zona CG terbang
3. **Jangan melebihi 7.09 cm** dari LE MAC (aft limit baru 0.96 ft)
4. Seimbangkan di dua jari → validasi dengan hand-launch glide

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4. Geometri Sayap

Wing 1

Parameter	Imperial	Metrik
Semi-length	**2.13 ft**	**64.92 cm**
Wingspan	**4.26 ft**	**~129.8 cm**
Root chord	0.78 ft	23.77 cm
Tip chord	0.60 ft	18.29 cm
Taper ratio	—	0.769
MAC	0.690 ft	21.03 cm
Wing area	2.94 ft²	**0.2731 m²**
Aspect ratio	—	**6.174**
Sweep / Dihedral	0.0° / 0.0°	—
Long arm (aero ref)	0.90 ft	27.43 cm

Wing 2 — Winglet

Parameter	Nilai
Semi-length	0.43 ft
Root chord	0.60 ft
Tip chord	0.08 ft
Sweep	80.0°
Dihedral	20.0°
Lat arm (= Wing 1 semi)	**2.13 ft**

Vertical Stabilizer — V-tail (×2)

Parameter	Imperial	Metrik
Height	0.65 ft	19.8 cm
Root chord	0.56 ft	17.1 cm
Tip chord	0.25 ft	7.6 cm
Sweep	40.0°	—
Dihedral kedua fin	45.0°	—

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5. Airfoil NACA 2412

Karakteristik Airfoil NACA 2412

Profil	**NACA 2412**	Camber 2% · max camber di 40% chord · t/c 12%
CLmax (Re~700k)	**1.50**	vs generic ~1.30 (+15%)
CD0 profil	**0.0075**	Lebih rendah dari foil tebal — laminar sebagian
CD0 total	**0.026**	vs generic 0.028 (–7.1%)
Oswald efficiency e	**0.82**	vs generic 0.78 (+5.1%)
L/D max	**12.37**	vs generic 10.7 (+15.6%)
Zero-lift AoA	**–2.1°**	Positif camber → slight nose-down at zero lift
Stall AoA	**~16.5°**	Stall ringan — gradual, forgiving
Cm_ac	**–0.05**	Nose-down pitching moment → CG tidak boleh terlalu belakang
AoA @ cruise	**3.83°**	Dari zero-lift + CL_cruise/Cl_alpha

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6. Performa Aerodinamis (MTOW 2.381 kg, NACA 2412)

Parameter	Nilai
Wing loading	**85.5 N/m²**
Stall speed Vs	**9.65 m/s (34.7 km/h)**
Min safe airspeed (1.2×Vs)	11.58 m/s (41.7 km/h)
Best cruise speed	**14.74 m/s (53.1 km/h)**
AoA saat cruise	**3.83°**
Best range speed	~16.95 m/s (61.0 km/h)
VNE	**33.77 m/s (121.6 km/h)**
L/D max	**12.37**
Thrust @ cruise	1.89 N

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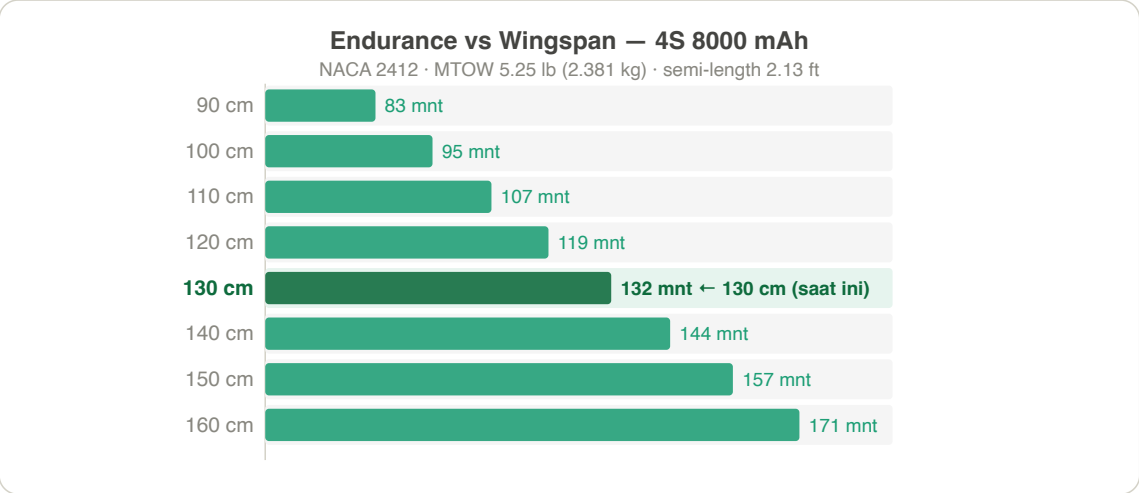
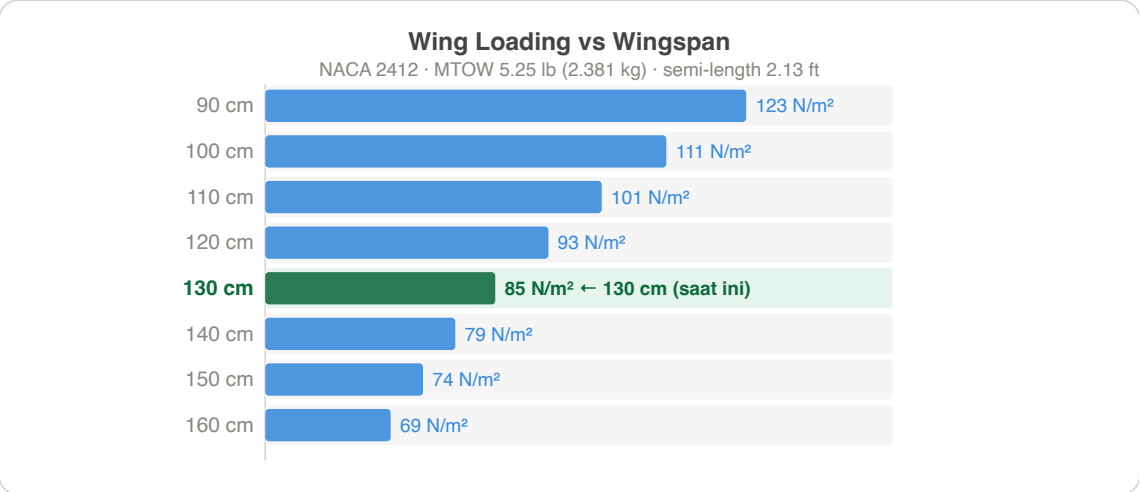
7. Analisis Wingspan vs Daya Angkat

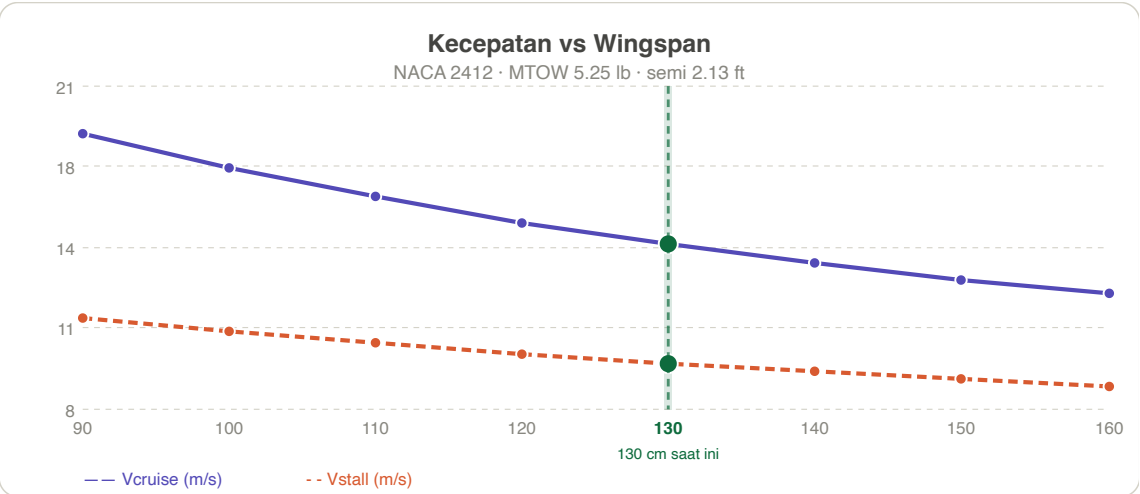
Endurance vs Wingspan



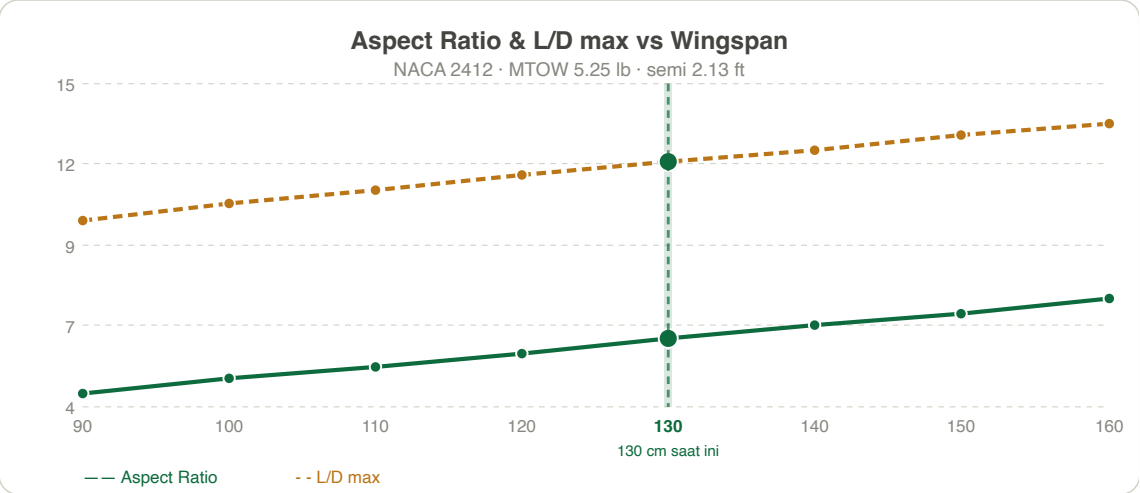
Wing Loading vs Wingspan



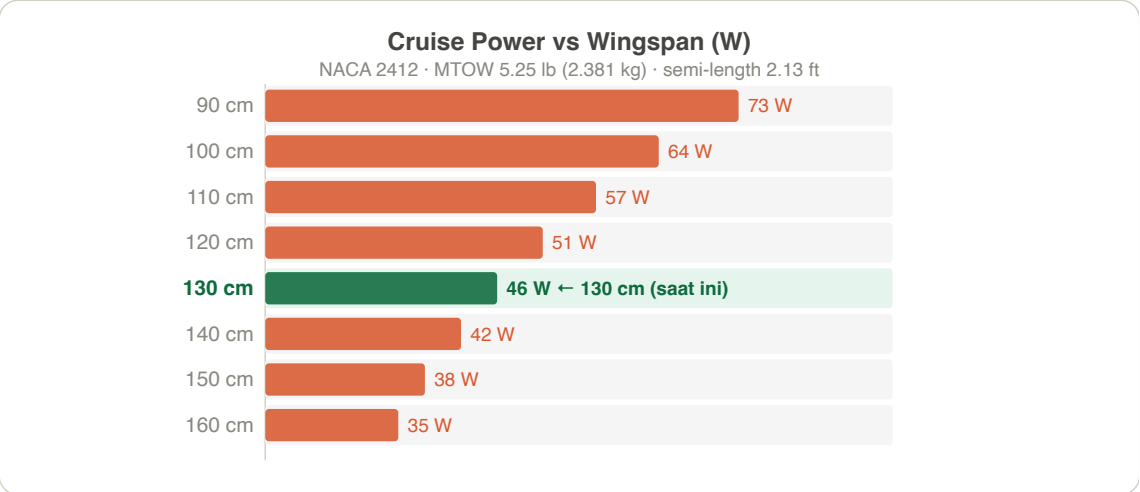
Kecepatan vs Wingspan



Aspect Ratio & L/D vs Wingspan



Cruise Power vs Wingspan



Tabel Lengkap

Wingspan	Area	AR	WL	Vs	Vcruise	L/D	P cruise	Endurance (4S 8Ah)
90 cm	0.189 m²	4.3	123 N/m²	11.6 m/s	19.4 m/s	10.3	73 W	83 mnt
100 cm	0.210 m²	4.8	111 N/m²	11.0 m/s	17.9 m/s	10.9	64 W	95 mnt
110 cm	0.232 m²	5.2	101 N/m²	10.5 m/s	16.7 m/s	11.4	57 W	107 mnt
120 cm	0.253 m²	5.7	93 N/m²	10.0 m/s	15.6 m/s	11.9	51 W	119 mnt
~130 cm ★ saat ini	**0.273 m²**	**6.17**	**85 N/m²**	**9.65 m/s**	**14.74 m/s**	**12.37**	**46 W**	**131 mnt**
140 cm	0.295 m²	6.7	79 N/m²	9.3 m/s	13.9 m/s	12.8	42 W	144 mnt

150 cm	0.316 m²	7.1	74 N/m²	9.0 m/s	13.2 m/s	13.3	38 W	157 mnt
160 cm	0.337 m²	7.6	69 N/m²	8.7 m/s	12.6 m/s	13.7	35 W	171 mnt

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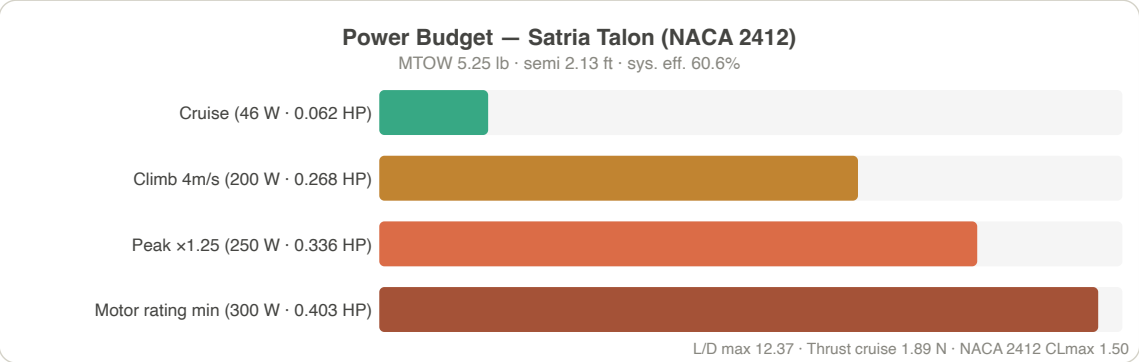
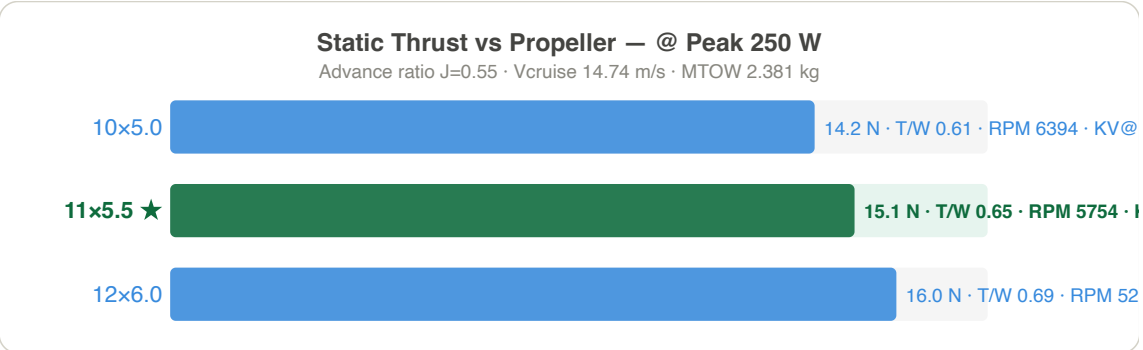
8. Propulsi

Power Budget



Kondisi	Daya Listrik	HP
Cruise (14.74 m/s)	**46 W**	**0.062 HP**
Climb (ROC 4 m/s)	200 W	0.268 HP
Peak + gust (×1.25)	**250 W**	**0.335 HP**
Motor rating min	**≥ 300 W**	**≥ 0.403 HP**

Propeller



Propeller	RPM cruise	KV @ 4S	T static	T/W
10×5.0	6.394	617	14.2 N	0.61
11×5.5 ★	**5.754**	**555**	**15.1 N**	**0.65**
12×6.0	5.275	508	16.0 N	0.69

Motor & ESC & Baterai

Parameter	Nilai
Motor	Brushless 2216, **KV 555–600 @ 4S**
Power rating	≥ 300 W (gunakan 350–450 W)
ESC	**40 A BLHeli_32**, BEC 5V/3A
Baterai rekomendasi	**4S 8000 mAh**
Cruise current	~~3.1 A (~~46 W)
C-rate saat peak	**2.1C** (sangat aman)

Baterai	Endurance	Kargo bersih	C-rate
3S 5000 mAh	~62 mnt	740 g	4.5C
4S 5000 mAh	~82 mnt	700 g	3.4C
4S 6000 mAh	~99 mnt	670 g	2.8C
4S 8000 mAh	**~131 mnt**	**571 g**	**2.1C**

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9. Misi Delivery

Kargo	Endurance	Vcruise	Range one-way	Radius RTH
300 g	~131 mnt	14.74 m/s	**~116 km**	**~49–52 km**
500 g	**~131 mnt**	**14.74 m/s**	**~116 km**	**~49–52 km**
571 g (maks)	~131 mnt	14.74 m/s	~116 km	**~49–52 km**

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10. Ringkasan Spesifikasi Final

Parameter	Nilai
Airfoil	**NACA 2412**
Semi-length	**2.13 ft**
Wingspan	**~129.8 cm**
Wing area	0.2731 m²
Aspect ratio	6.174
MAC	21.03 cm · 0.690 ft
Taper ratio	0.769

Konfigurasi ekor	V-tail twin fin (dihedral 45°)
Empty weight	**3.00 lb = 1.361 kg**
MTOW	**5.25 lb = 2.381 kg**
Payload total	1.021 kg
Kargo (4S 8Ah)	**571 g**
CG forward limit	**0.91 ft = 26.4% MAC = 5.56 cm dari LE MAC**
CG nominal (terbang)	**0.92 ft = 27.9% MAC = 5.87 cm dari LE MAC ✓**
CG aft limit	**0.96 ft = 33.7% MAC = 7.09 cm dari LE MAC**
Wing loading	85.5 N/m²
Stall speed Vs	**9.65 m/s (34.7 km/h)**
Cruise speed	**14.74 m/s (53.1 km/h)**
AoA cruise	3.83°
VNE	33.77 m/s (121.6 km/h)
L/D max	**12.37**
Cruise power	**46 W (0.062 HP)**
Peak power	250 W (0.335 HP)
Motor rating min	300 W (0.403 HP)
Endurance (4S 8Ah)	**~131 mnt**
Range one-way	**~116 km**
Radius RTH	**~49–52 km**
Motor	2216, **KV 555–600 @ 4S**
Propeller	**APC 11×5.5E**
ESC	40 A BLHeli_32
Baterai	**4S 8000 mAh**

UAV Satria Talon · NACA 2412 · Semi 2.13 ft · V-tail · Empty 3.00 lb · MTOW 5.25 lb
CG updated: fwd 0.91 ft · nominal 0.92 ft · aft 0.96 ft (diperketat untuk NACA 2412)
Seluruh nilai performa adalah estimasi teoritis — wajib divalidasi melalui uji terbang.